

Fertility

<!-- SCAFFOLD: write order = Hook -> Female Fertility -> Male Fertility -> Conception -> Infertility -> Fertility in Aging. Reading level: <= 9th grade. Biology light, psychology heavy. Hormones folded into Female/Male sections (no standalone hormones section). Citations come from the two source chapters (Hyde Ch. 6; "TH" Ch. 11). Build a running References list at the bottom. -->

What Do You Already Know About Your Body?

Quick quiz. True or false?

- You can't get pregnant during your period.
- The "fertile window" covers most of the month, so timing is easy.
- PMS is mostly in a person's head.
- Menopause happens when a woman's hormones simply run out.
- Men make sperm fresh, on demand, in endless supply.

If you answered "true" to any of these, you're in good company. Most people do. You're also wrong on every single one.

That's not meant as an insult. It's kind of the point. Most of us pieced together what we "know" about fertility from a rushed health class, a few awkward conversations, friends who were just as lost, and whatever the internet decided to show us. The result is a patchwork of half-truths, old wives' tales, and confident guesses.

This chapter is the repair job. We'll take the common beliefs about fertility — how bodies make eggs and sperm, how a pregnancy actually starts, why some people struggle to conceive, and what changes with age — and hold them up against what the science really says. Some of what you "know" will hold up. A lot of it won't.

One more thing worth saying up front: science doesn't have every answer here, and we'll be honest about that. On a few topics, researchers are still actively arguing. Learning where the solid ground ends and the open questions begin is part of thinking clearly about your own body. So treat this less like a list of facts to memorize and more like a tour of what we know, what we're still figuring out, and how to tell the difference.

Let's start where the cycle of life literally begins each month.

Female Fertility

<!-- WRITTEN BELOW: cycle (5 phases, with endometrium under luteal, prostaglandins under premenstrual, products+TSS under menses), tracking ovulation, psychology of menstruation (PMS/PMDD, sex+myths, cross-cultural). -->

The female reproductive system runs on a roughly monthly rhythm called the menstrual cycle. Before we follow that cycle, it helps to meet the small cast of chemical messengers that run the whole show: hormones.

Think of hormones as text messages sent through the bloodstream. The brain (two parts called the hypothalamus and the pituitary) sends out messages — first a starter signal, then two main hormones known as FSH and LH. These travel down to the ovaries and tell them what to do. The ovaries reply with their own messages, mainly two hormones called estrogen and progesterone. The back-and-forth works like a thermostat: the brain sends a signal, the ovaries respond, and that response tells the brain whether to keep sending or ease off. You don't need to memorize every name. Just remember the basic loop — brain talks to ovaries, ovaries talk back — because that conversation is what drives everything below (Silverthorn, 2013).

The Menstrual Cycle: A Monthly Reset

Here's the big idea before any details. Each month, the body gets ready for a possible

pregnancy. It builds a soft, blood-rich lining inside the uterus — called the endometrium — like preparing a nursery for a guest who might arrive. If a fertilized egg shows up and settles in, the nursery is ready. If no one comes, the body clears out the lining and starts fresh. That clear-out is the period.

So a period isn't a malfunction or a sign something's wrong. It's the reset button.

A quick note on counting: doctors call the first day of bleeding "day 1," simply because it's the easiest moment to spot. The cycle averages about 28 days, but anywhere from about 21 to 35 days is common and perfectly normal. With that in mind, let's walk through the five phases.

The Five Phases

1. Follicular phase. In the ovaries, a handful of tiny fluid-filled sacs called follicles begin to grow, each holding one immature egg. Picture a few seeds sprouting at once — usually only one outgrows the rest and keeps maturing. Meanwhile, rising estrogen starts rebuilding the uterine lining. This phase varies in length more than any other, which is the main reason cycles differ from person to person and even month to month.

2. Ovulation. The one mature egg bursts out of its follicle and gets swept into the fallopian tube. This is the most fertile moment of the whole cycle, landing somewhere around the middle — roughly day 14 in a 28-day cycle, though it slides earlier or later for many people. The body gives off several signals around this time, which we'll use in the next section.

3. Luteal phase. The empty follicle left behind doesn't just disappear. It becomes a small, temporary gland called the corpus luteum that pumps out progesterone. Progesterone's job is to keep that uterine lining plush and ready, and to keep the uterus calm and quiet. This is the steady, predictable half of the cycle — it's almost always about 14 days long. (So when cycles run long or short, it's the *first* half doing the varying, not this one.) This is also where

the endometrium sits at its richest and most welcoming, waiting to see if a fertilized egg will arrive and settle in.

4. Premenstrual phase (the "late luteal" phase). If no fertilized egg implants, the corpus luteum shuts down, and estrogen and progesterone both drop. That drop matters. With progesterone gone, chemicals called prostaglandins rise. Prostaglandins are basically a "contract now" signal — they make the muscle of the uterus squeeze. Those squeezes are what we feel as cramps. Bloating, fatigue, and mood changes can show up here too. (This is also why a medicine like ibuprofen helps with cramps: it lowers prostaglandins, turning down the "contract" signal.)

5. Menses (the period). Now hormones are at their lowest. Without estrogen and progesterone to hold it in place, the built-up lining breaks down and leaves the body through the vagina over a few days. Then hormone levels begin to rise again, the next batch of follicles starts growing, and the whole cycle restarts.

A practical note that belongs right here: people manage menstrual flow in several ways — pads, tampons, menstrual cups, period underwear, and menstrual discs. Each has trade-offs in cost, comfort, convenience, and environmental impact, and there's no single "right" choice. One safety point is worth knowing: tampons and cups left in too long can, in rare cases, lead to toxic shock syndrome (TSS), a serious infection. That's why following the timing instructions on the package — changing tampons and emptying cups on schedule — actually matters.

Tracking Ovulation: Reading Your Body's Signals

Your body doesn't keep ovulation a secret. It gives off signals around that fertile window, and learning to read them is useful for all kinds of reasons — whether someone is trying to get pregnant, trying to avoid it, or just wants to understand their own rhythm. Here are the main methods, from least to most reliable.

- **The calendar method.** Count forward from the start of the cycle and assume ovulation lands around the middle. It's the simplest method, but also the least reliable, because ovulation day moves around. Despite the "day 14" rule of thumb, most people don't ovulate exactly on day 14.
- **Cervical mucus.** Near ovulation, cervical mucus changes from dry and sticky to clear, slippery, and stretchy — a lot like raw egg white. That slippery mucus actually helps sperm move, and it's a surprisingly good signal that the fertile window is open.
- **Cervical changes.** Around peak fertility, the cervix itself softens and opens slightly. People who learn to check can use this alongside the mucus signal.
- **Basal body temperature (BBT).** Your at-rest body temperature ticks up about half a degree right after ovulation (that's progesterone's doing). The catch: it only confirms ovulation *after* it has already happened, and it has to be taken first thing in the morning, before getting out of bed, or the reading is thrown off.
- **Ovulation predictor kits (OPKs).** These drugstore tests detect the surge of LH that comes roughly a day before ovulation. They're precise and easy, but repeated testing can get expensive — and stressful.

One fact ties all of this together and clears up a common misunderstanding about the fertile window. The egg itself can only be fertilized for about 12 to 24 hours after it's released. But sperm are tougher: they can survive inside the body for up to about five days (Wilcox et al., 1995). Add those together and the truly fertile window is only several days long — not most of the month, the way many people assume. (Hold onto that five-day sperm survival fact. It's about to explain one of the biggest myths in this whole chapter.)

The Psychology of Menstruation

Here's where a human sexuality course really digs in, because a period is never *just* biology. It comes wrapped in mood, meaning, relationships, and culture. The same physical event can feel like a burden in one home and a celebration in another — and a big part of why comes down to what people are taught to believe about it.

PMS (and Its Severe Cousin, PMDD)

Premenstrual syndrome, or PMS, refers to the physical and emotional changes that show up in the week or two before a period (Yonkers et al., 2008). The list is familiar to a lot of people: mood swings, irritability, sadness, anxiety, bloating, fatigue, tender breasts, cramps, and food cravings. The main engine behind it is that late-cycle hormone drop — falling estrogen and progesterone, plus rising prostaglandins (Bäckström et al., 2003).

But here's the part that surprises students: PMS is far less dramatic, on average, than the culture makes it sound. Up to about 80% of people who menstruate notice at least one or two symptoms that are annoying but not disruptive. Around 20 to 30% of college-age women report more moderate discomfort. And only about 3 to 8% have symptoms severe enough for an actual clinical diagnosis, called premenstrual dysphoric disorder (PMDD) — a condition that is real, treatable, and shouldn't be brushed off (Biggs & Demuth, 2011; Bäckström et al., 2003).

So why does PMS loom so large in our imagination? Partly because the media has long exaggerated it, which fed a lazy and sexist habit of waving off any woman's bad mood with "oh, she's PMS-ing." That move does double damage: it dismisses real feelings *and* blames the body. Culture can also make the physical side feel worse. When a society ties a woman's worth to looking attractive and "clean," the bloating and blemishes of PMS land harder than they otherwise would. Some researchers go further, arguing that negative cultural messages don't just amplify PMS — they help create it, as people absorb the idea that their bodies are a problem.

Sex, Periods, and the Myths Around Them

In much of Western culture, there's an unspoken rule that periods are gross, shameful, or somehow "unclean." That belief — call it the menstrual taboo — shapes real behavior. People may avoid sex during their period, feel less attractive, or feel embarrassed about the

whole thing. Partners may react with discomfort or avoidance, sometimes simply because no one ever taught them otherwise.

A reality check helps. Menstrual blood is sterile on its own and isn't a health hazard. There are a couple of small considerations during a period — the cervix is slightly open, which makes it a little easier for infections to travel upward — but there's also a perk worth mentioning: orgasm can actually ease cramps. Couples who want to have sex during a period manage flow in simple ways, like timing, a towel, or a menstrual cup.

Now for the big one — maybe the most stubborn myth in this chapter: *"You can't get pregnant during your period."* It's false. Remember that sperm can survive up to about five days, and that cycle length varies a lot. In a shorter cycle, ovulation can arrive soon after the period ends. So sperm deposited during period sex can still be alive and waiting when an egg is released a few days later (Wilcox et al., 1995). The odds are lower than at peak fertility, sure — but "lower" is not "zero." Anyone using this as their birth control plan is rolling the dice.

What Periods Mean Across Cultures

Step back from any one culture and a striking pattern appears: the biology of menstruation is the same everywhere, but the *meaning* people attach to it swings wildly. One useful way to map that range is a spectrum from erotophobic (fearful and negative about sex and the body) to erotophilic (positive and celebratory). Attitudes tend to fall into four broad groups.

- **Very erotophobic.** Menstruation is treated as contamination — "the curse." People who are menstruating may be separated, hidden, or treated as if they're diseased. Among the Dogon people of Mali, for example, menstruating women spend about five days each month in separate huts (Strassmann, 1992). Researchers suspect the huts may serve a hidden purpose: letting men track exactly when their wives menstruate. Strikingly, genetic testing found that Dogon fathers were far less likely to be unknowingly raising another man's child than men in nearby communities who

didn't follow the practice (Strassmann et al., 2012). These taboos can also turn deadly — in 2017, a girl in Nepal died from a snakebite while isolated alone in a menstrual hut (Pokharel, 2017).

- **Erotophobic (milder).** Menstruating people are seen as unappealing or "off." This is common in many places, including parts of the United States, where shame and embarrassment still linger around periods.
- **Erotophilic.** A first period is welcomed as a passage into adulthood, often marked with gifts or a celebration.
- **Very erotophilic.** Menstruation is viewed as sacred or even healing.

The celebratory end of the spectrum is easy to overlook if you grew up around the taboo. In some Japanese families, a first period is honored with *sekihan*, a red bean and rice dish whose color stands for happiness. The Navajo *Kinaalda* is a four-day coming-of-age ceremony that includes running to show strength. Among the Nuu-chah-nulth of the Pacific Northwest, a girl's first period is celebrated as she swims from a boat back to shore while her village cheers her on. In these traditions, menstruation signals strength and arrival, not shame.

The takeaway for a psychology course is this: a teenager's body does the same thing in Mali, Tokyo, and Ohio. What differs is the story told about it — and that story shapes how a person feels about their own body for years to come.

Male Fertility

<!-- DETAILED OUTLINE / GAME PLAN — not final prose. To be written at <=9th grade, biology light, psychology heavier, with metaphors. Inline **[SRC]** = factual claim that will need a scientific source. Most of this section draws on outside sources, not the two provided chapters. -->

Opening framing (parallel to the Female Fertility hormone intro): the big contrast is rhythm. The female system runs on a ~monthly loop; the male system runs more or less continuously —

a factory that never closes. Reuse the "text-message / thermostat" hormone metaphor so this section doesn't depend on the deleted hormones chapter.

The Male Hormone System (folded in, kept light)

- The key difference: non-cyclical, continuous production. Metaphor: female system = monthly event planner; male system = 24/7 factory. [SRC]
- The loop, light: brain sends GnRH → pituitary sends FSH + LH → testes respond. LH → Leydig cells make testosterone; FSH (with testosterone) → Sertoli cells run sperm production; inhibin = the "ease off, we have enough" feedback signal. [SRC]
- Testosterone does more than make sperm (sex drive, body changes) — one-line mention, don't over-develop here. [SRC]
- Relatable fact: testosterone follows a daily rhythm (higher in the morning) and shows some seasonal variation. [SRC]
- *Possible psychology beat (see Q3): a quick myth-check on "testosterone = aggression/dominance," which is far more oversimplified than pop culture suggests.* [SRC]

Sperm Production

A. Spermatogenesis — how sperm are made (biology light)

- Where it happens: seminiferous tubules. Sertoli cells = "nurse cells" that feed and protect developing sperm; Leydig cells = the testosterone factory; the blood-testis barrier = a "security wall" that hides sperm from the body's own immune system. Metaphor: a protected nursery. [SRC]
- The progression in one light line (stem cell → ... → mature sperm); do NOT wade into meiosis. [SRC]
- The hook — scale + speed: each sperm takes ~64-74 days to finish, but the line never stops, cranking out on the order of hundreds of millions a day (~1,000+ per second). Metaphor: a 2.5-month assembly line running nonstop, so finished product rolls off constantly. [SRC]

- Why testes hang outside the body: sperm-making needs to run a few degrees cooler than core temperature. Lifestyle tie-in (heat from hot tubs, saunas, laptops) → forward link to section D. [SRC]
- Compare/contrast hook (see Q5): people with ovaries are born with all their eggs; people with testes make brand-new sperm nonstop for life. Sets up the aging section later. [SRC]

B. The journey to the egg — beyond the "race" myth (CENTERPIECE)

- Reframe the cultural image: millions of sperm sprinting, the strongest/fastest "winning" and forcing its way in. Mostly wrong. [SRC]
- The numbers: ~200–300 million start; only a tiny handful ever get near the egg; massive attrition. [SRC]
- The female tract is active, not a passive racetrack — it filters, guides, stores, and may select (bridge to C). [SRC]
- Capacitation: sperm aren't "ready" on arrival; the female tract has to switch them on before they can fertilize. Metaphor: reaching the trailhead but needing to gear up first. [SRC]
- The three guidance systems (organizing device): rheotaxis (orienting/swimming against fluid flow — like a fish facing upstream); thermotaxis (following a temperature gradient toward the warmer site); chemotaxis (homing in on chemical signals near the egg, e.g., progesterone — like following a scent). [SRC]
- *Psychology / sociology-of-science beat (see Q1): the "heroic warrior sperm vs. passive princess egg" story shaped even how scientists described reproduction — Emily Martin's classic critique. Strong critical-thinking hook.* [SRC]

C. Gamete selection and "cryptic female choice" (EMERGING — flag as contested)

- The idea: which sperm succeeds may not be random or simply "fastest" — the female tract/egg may tilt the odds. Term: cryptic female choice. [SRC]

- Human hint: the tract may favor sperm from immune-gene-dissimilar (HLA-dissimilar) partners — ties to mate choice and the biology of "chemistry." [SRC]
- *Optional attraction link (see Q2): the MHC "sweaty T-shirt" attraction studies. Popular, but mixed/failed replications — include only with caveats.* [SRC]
- Mechanism: female reproductive fluid can attract and favor certain/better sperm; possible implications for assisted reproduction. [SRC]
- Heavy caveat to model good science: much evidence is from non-human species; human findings are preliminary. Teach as "frontier," not fact. [SRC]

D. Sperm count and quality — the decline debate (critical-thinking / media-literacy)

- What a semen analysis measures: count, concentration, motility, morphology (define each in plain words). [SRC]
- The headline: Levine et al. meta-analyses — average sperm count roughly halved since the 1970s (~104M → ~49M/mL, 1973–2019), reportedly accelerating, now reported worldwide. [SRC]
- Proposed drivers (sole home for this material; cross-ref from 5.2): endocrine-disrupting chemicals in plastics (phthalates, BPA), obesity, smoking, inactivity, ultra-processed diets, heat. [SRC]
- The pushback (the teachable part): possible measurement artifact; the "biovariability" critique; declines may be regional/cohort-specific rather than uniform; "normal" spans a wide range. [SRC]
- *Psychology / society beat: why does the scary version spread? Media amplification, "spermageddon" framing, fertility anxiety, masculinity. Good class-debate + media-literacy material.* [SRC]

Closing: short bridge into Conception (the sperm has arrived and been activated — now what actually happens at the egg).

Claims to source (Male Fertility)

Established physiology (textbook-level; Silverthorn 2013 covers much of it, plus one or two physiology/clinical refs):

- Seminiferous tubules / Sertoli ("nurse") + Leydig (testosterone) cells / blood-testis barrier
- Male hormone loop (GnRH → FSH/LH → testosterone + spermatogenesis; inhibin feedback); continuous (non-cyclical) production
- Spermatogenesis takes ~64-74 days
- Output ~hundreds of millions/day
- Testes external for cooler temperature
- Testosterone's daily/seasonal rhythm
- Born-with-all-eggs vs. lifelong sperm production contrast
- ~200-300 million sperm per ejaculate; heavy attrition

Specialized but solid (sperm navigation — from reproductive-biology literature):

- Guided, not coincidental, arrival; female tract filters/guides/stores
- Capacitation as a required activation step
- Rheotaxis / thermotaxis / chemotaxis (progesterone as chemoattractant)

Emerging / contested (require explicit caveats):

- Cryptic female choice; female tract favoring HLA-dissimilar sperm
- Female reproductive fluid favoring better sperm
- (If used) MHC "sweaty T-shirt" attraction studies — note mixed replication
- Emily Martin's gendered-language critique (humanities/STS source, not empirical)

Decline debate (need balanced sourcing — present both sides):

- Levine et al. (2017, 2022) decline figures
- Proposed environmental/lifestyle drivers

- The methodological / "biovariability" counter-critique

<!-- OPEN QUESTIONS FOR ROSS (also in chat): Q1 — Include the Emily Martin "Egg & the Sperm" sociology-of-science critique? How prominent? Q2 — Include the MHC "sweaty T-shirt" attraction studies in section C, with caveats? Or keep C strictly to cryptic female choice? Q3 — Want a testosterone-and-behavior myth-check here, or save all behavior/libido for another chapter? Q4 — Decline debate: full class-debate centerpiece (longer, both sides at length) or a tighter treatment? Q5 — Draw the eggs-vs-sperm contrast here (sets up aging), or keep the male and female sections cleanly separate? Q6 — Male Fertility will run shorter than Female Fertility unless we build up the psychology angles (science-narrative critique, fertility anxiety, masculinity). Build them up for balance, or let it run shorter? -->

Conception

(to be written — sperm structure and the male tract, fertilization mechanics)

Infertility

(to be written)

Fertility in Aging

(to be written)

References

(running list — added as sections are written)

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